

D 4 s later

2 What is a reasonable estimate of the current in an electric kettle that is in use?

A $8\mu\text{A}$

B 8 mA

C 8 A

D 8 kA

3 Which statement about errors in measurements is correct?

A An accurate set of measurements always has a small random error.

B A precise set of measurements always has a small systematic error.

C A random error can be reduced by taking an average of several measurements.

D A systematic error creates a random set of measurements spread out about the true value.

4 A sample of material has cross-sectional area A and length L . The temperatures at the two sides of the sample are T_1 and T_2 . Thermal energy Q is transferred through the sample in time t .

These quantities are related by

$$\frac{Q}{t} = \frac{k \times A \times (T_1 - T_2)}{L}$$

where k is a constant.

What are the SI base units of k ?

A $\text{kg m s}^{-3} \text{ } ^\circ\text{C}^{-1}$

B $\text{kg m s}^{-3} \text{ K}^{-1}$

C $\text{kg m s}^{-1} \text{ } ^\circ\text{C}^{-1}$

D $\text{kg m s}^{-1} \text{ K}^{-1}$